

CUSTOMER CHURN PREDICTION

Predicting At-Risk Telecom Customers with Machine Learning

Project Documentation

Advanced Data Analysis Course

Team Members: [Sawsan Sayed , Doaa Mohamed , Mohamed Ihab , SamehAli]

Instructor: [Hagar Hisham]

Submission Date: [18/9/2026]

Table of Contents

Table of Contents	2
1. Executive Summary	4
2. Problem Statement	4
The question	4
Why it matters	4
Scope and boundaries	4
3. Data Sources	5
4. Data Architecture / Data Schema	5
Field groups	5
Data dictionary	5
5. Data Cleaning & Preparation	6
Before / after snapshot	7
6. Exploratory & Statistical Analysis	7
Descriptive statistics	7
Key patterns	7
What these patterns suggested	8
7. Technique Application — Modeling	10
Track chosen: Predictive / Classification	14
Feature preparation	14
Models tried and final choice	15
Final model evaluation (held-out test set)	15
Streamlit app showcase	15
8. Dashboard & Inferred Insights	21
Dashboard structure	21
Key insights	21
9. Conclusion	22
Answer to the original question	23
Limitations	23
Recommendations / next steps	23
10. Appendices	24
Links	24
Data source citation	24
Supplementary materials	Error! Bookmark not defined.

1. Executive Summary

Business question: Can we identify, in advance, which telecom customers are likely to churn — so the retention team can intervene before they leave?

Key finding: Churn is concentrated among customers on month-to-month contracts, paying by electronic check, with shorter tenure and higher monthly charges; churn rate also varies meaningfully by city (~24%–31%).

Recommended action: Deploy the trained classification model (delivered as a live Streamlit app) to score customers in real time, and route customers flagged as high-risk to the retention team for proactive outreach — for example, a targeted offer before a month-to-month contract renews.

Final model performance on the held-out test set: [insert Accuracy / Precision / Recall / F1 / ROC-AUC from your notebook's Step 8 output].

2. Problem Statement

The question

Telecom providers lose a significant share of their subscriber base every year — in this dataset, 26.5% of customers have churned. By the time a customer formally cancels, the opportunity to retain them has already passed. This project asks: can we predict, ahead of time, which customers are at high risk of churning, using information the company already has on file?

Why it matters

- Acquiring a new customer typically costs 5–25× more than retaining an existing one, so even a modest reduction in churn has a direct impact on revenue.
- The retention team benefits directly: a ranked list of at-risk customers lets them prioritize outreach (discounts, service fixes, check-ins) instead of treating every customer the same.
- Company leadership benefits from visibility into why customers leave — the contract types, payment methods, and service gaps most associated with churn — which can inform product and pricing decisions beyond just the model itself.

Scope and boundaries

- The dataset is a single snapshot of 7,042 customers — it does not track how a given customer's risk changes over time (no repeated observations per customer).
- The analysis covers residential telecom customers across 13 cities (after data cleaning — see Section 5) in the source data; the model's conclusions should not be assumed to generalize to markets or customer segments outside this dataset without validation.

- The project focuses on the classification task itself (will a customer churn) rather than downstream financial modeling of exact retention-campaign ROI.

3. Data Sources

Field	Detail
Dataset name	Telco Customer Churn (extended with a city field)
Origin / link	https://www.kaggle.com/code/basmalaawad/telco-customer-churn-dataset/notebook
Access date	13/9/2026
Size	7,042 customer records × 23 columns (after cleaning)
Granularity	One row per customer — a snapshot, not a time series
Licensing / usage notes	[insert license terms shown on the source page, e.g. Kaggle dataset license]

4. Data Architecture / Data Schema

The dataset is a single flat table — one row fully describes one customer at one point in time. There are no separate linked tables to join, so no ER diagram is required; the structure below groups the 23 columns by what they describe.

Field groups

- Customer profile: gender, senior_citizen, partner, dependents, city
- Subscribed services: phone_service, multiple_lines, internet_service, online_security, online_backup, device_protection, tech_support, streaming_tv, streaming_movies
- Account & billing: contract, paperless_billing, payment_method, tenure, tenure_group, monthly_charges, total_charges
- Target: churn

Data dictionary

Field	Type	Meaning
customer_id	string	Unique customer identifier
city	categorical	Customer's city (13 categories after cleaning)
gender	categorical	Male / Female
senior_citizen	categorical	Whether the customer is a senior citizen (Yes/No)

Field	Type	Meaning
partner	categorical	Whether the customer has a partner
dependents	categorical	Whether the customer has dependents
tenure	numeric	Months the customer has stayed with the company
phone_service	categorical	Whether the customer has phone service
multiple_lines	categorical	Yes / No / No phone service
internet_service	categorical	DSL / Fiber optic / No
online_security	categorical	Yes / No / No internet service
online_backup	categorical	Yes / No / No internet service
device_protection	categorical	Yes / No / No internet service
tech_support	categorical	Yes / No / No internet service
streaming_tv	categorical	Yes / No / No internet service
streaming_movies	categorical	Yes / No / No internet service
contract	categorical	Month-to-month / One year / Two year
paperless_billing	categorical	Yes / No
payment_method	categorical	Electronic check / Mailed check / Bank transfer / Credit card
monthly_charges	numeric	Current monthly charge amount
total_charges	numeric	Total amount charged to the customer to date
tenure_group	categorical	Tenure bucketed into ranges (derived from tenure)
churn	categorical (target)	Whether the customer churned — Yes / No

5. Data Cleaning & Preparation

The table below documents every data-quality issue identified, how it was diagnosed, and the exact fix applied — full code for each step is in the accompanying notebook.

Issue	Diagnosis	Fix applied
TotalCharges stored as text with blank values	11 rows had a blank string instead of a number (new customers with tenure = 0, not yet billed)	Converted the column to numeric; resolved the blanks so every row has a valid numeric value
Duplicate customer records	Checked for exact duplicate rows and duplicate customer_id values	0 duplicates found — no rows removed on this basis

Issue	Diagnosis	Fix applied
Inconsistent city spelling	The same city appeared under two different spellings: "Faiyum" (414 rows) and "Faiyoum" (216 rows)	Standardized both to a single spelling ("Faiyum"), reducing the city category count from 14 to 13 and restoring the true churn-rate signal for that city
Non-predictive / redundant columns	customer_id is a unique identifier; tenure_group duplicates the information already in tenure	Dropped both before modeling (kept in the dataset for reference/dashboard use, excluded from the feature set)
Outlier check	Checked tenure, monthly_charges, and total_charges using the IQR method	No meaningful outliers found — all values fall within expected business ranges

Before / after snapshot

Metric	Before cleaning	After cleaning
Missing values (TotalCharges)	11 blank rows	0
Duplicate rows	0	0
Unique city categories	14 (with a spelling duplicate)	13
Feature columns used for modeling	23 raw columns	21 (customer_id and tenure_group dropped)
Churn class balance	26.5% Yes / 73.5% No	Unchanged (imbalance handled at modeling time via SMOTE, not by altering the raw data)

6. Exploratory & Statistical Analysis

Descriptive statistics

- 7,042 customers; overall churn rate 26.5% (imbalanced target).
- Tenure, monthly_charges, and total_charges show no significant outliers by the IQR method; monthly_charges and total_charges are right-skewed, consistent with a mix of low-usage and high-usage customers.

Key patterns

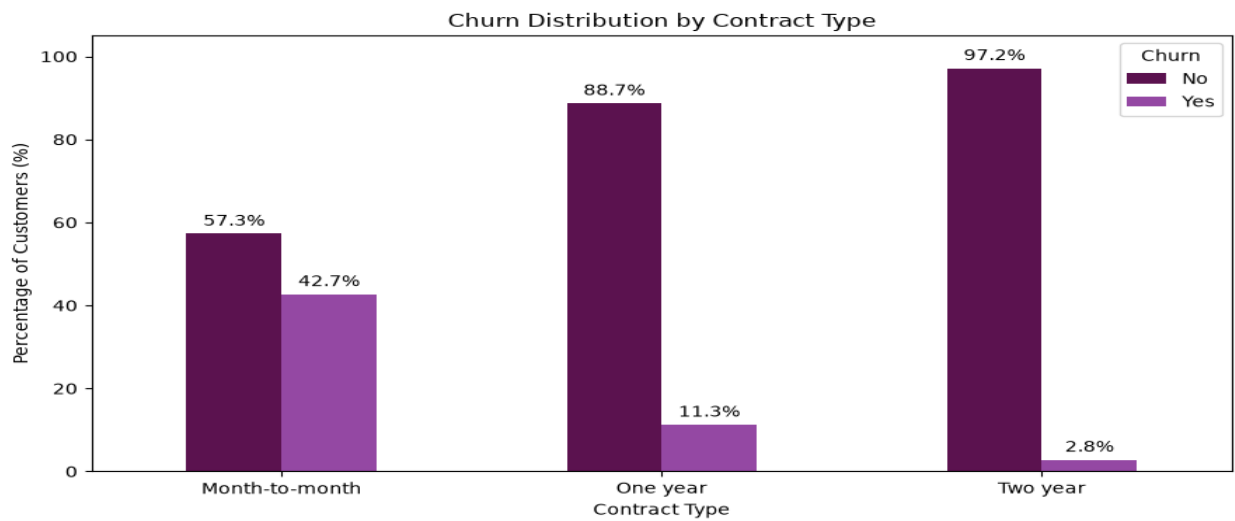
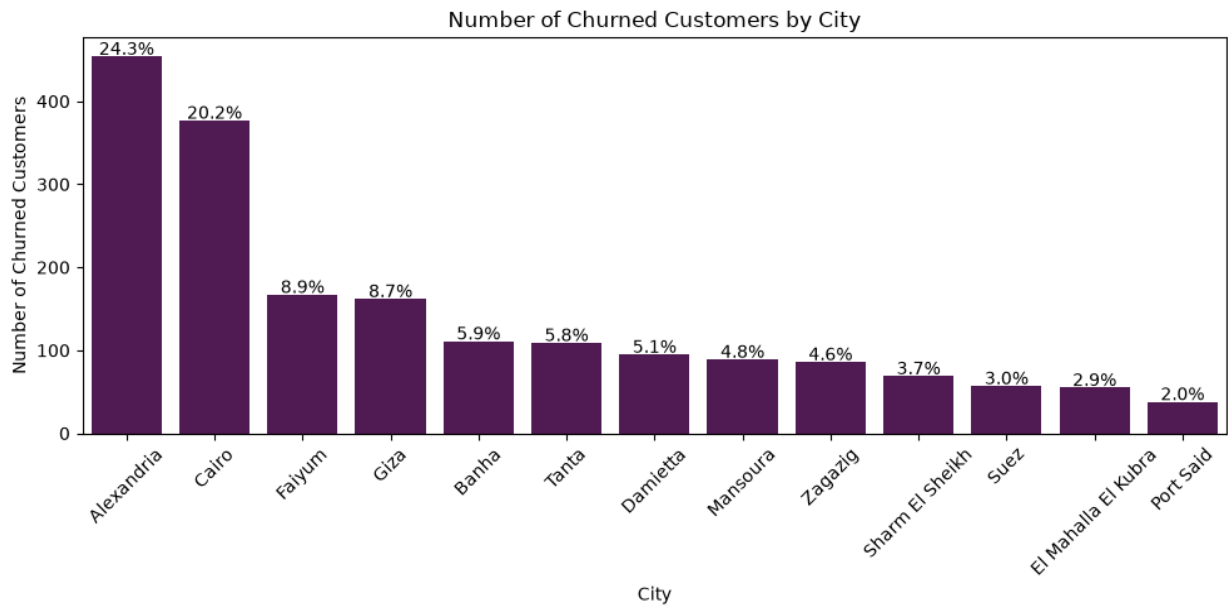
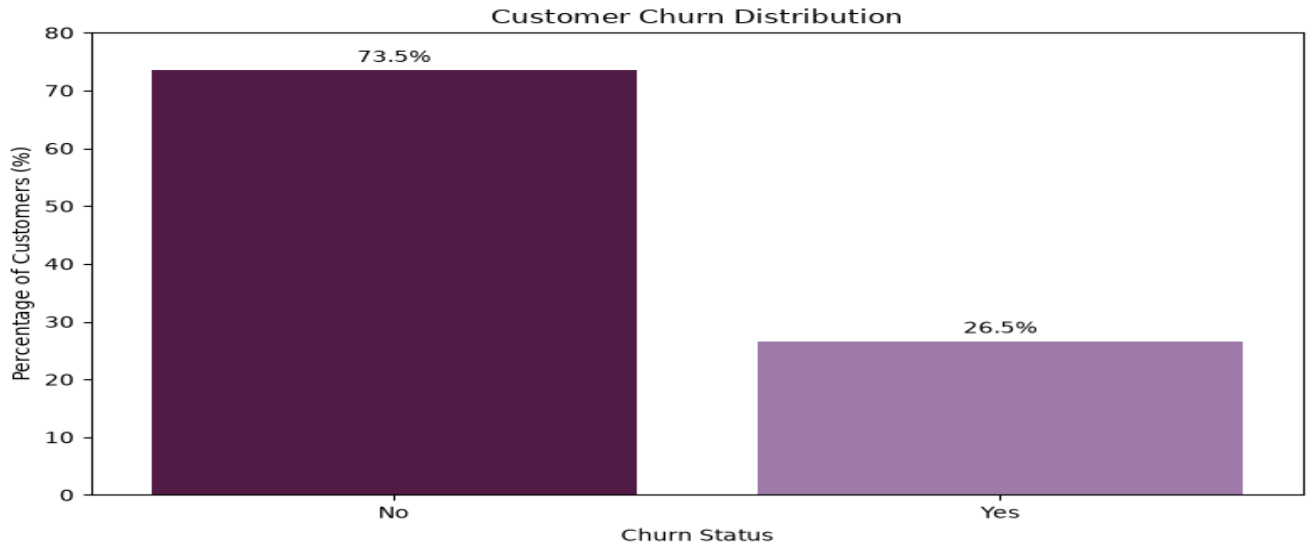
- Contract type: month-to-month customers churn at a markedly higher rate than one- or two-year contract holders — the single strongest categorical driver observed.
- Payment method: customers paying by electronic check show the highest churn rate of the four payment methods.
- Tenure and monthly charges: churned customers tend to have shorter tenure and higher monthly charges than retained customers.

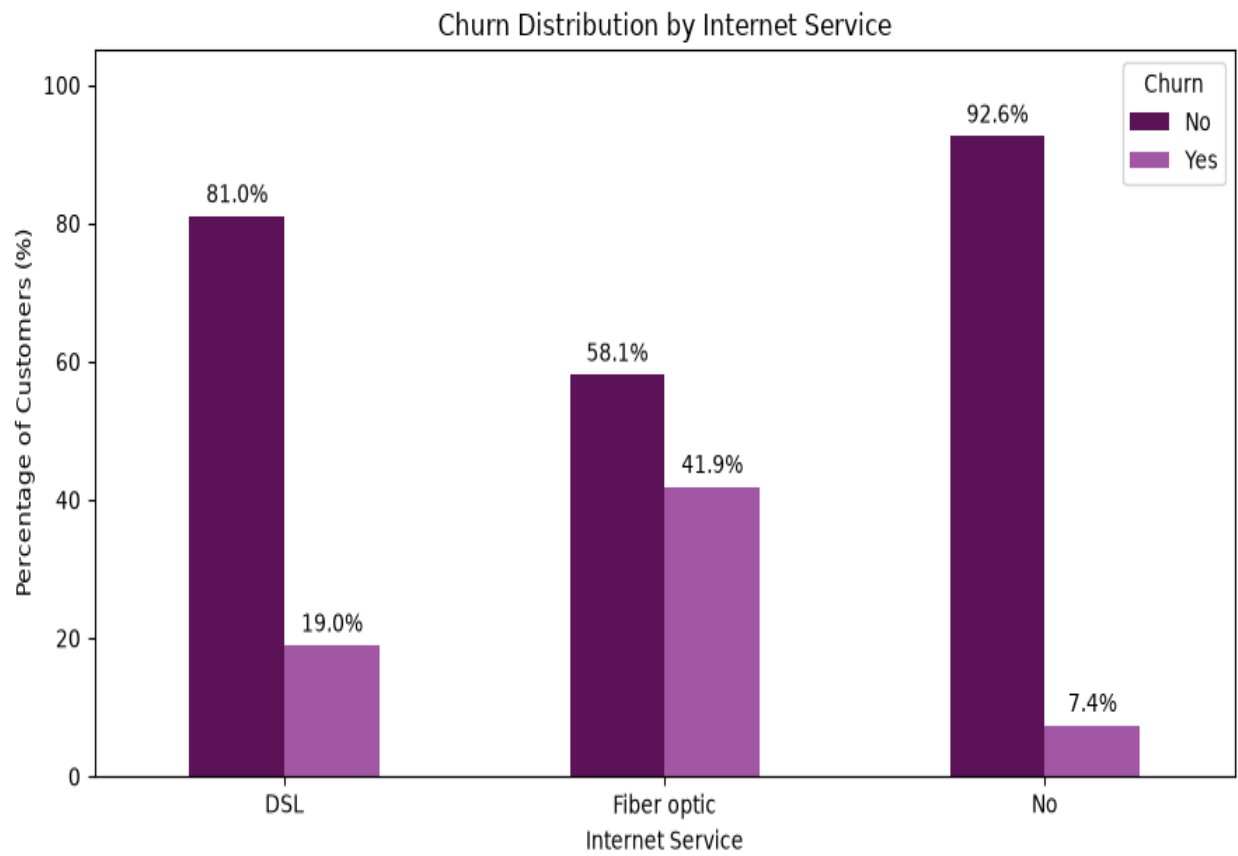
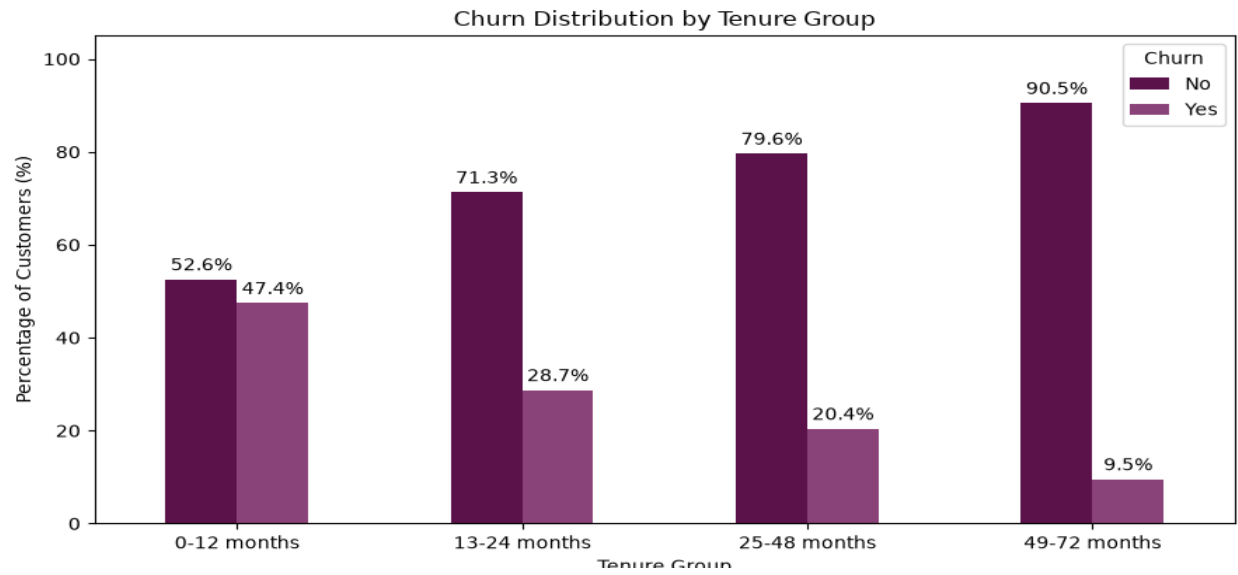
- City: churn rate ranges from about 24% (lowest) to about 31% (highest) across the 13 standardized cities — a real, if moderate, geographic effect.

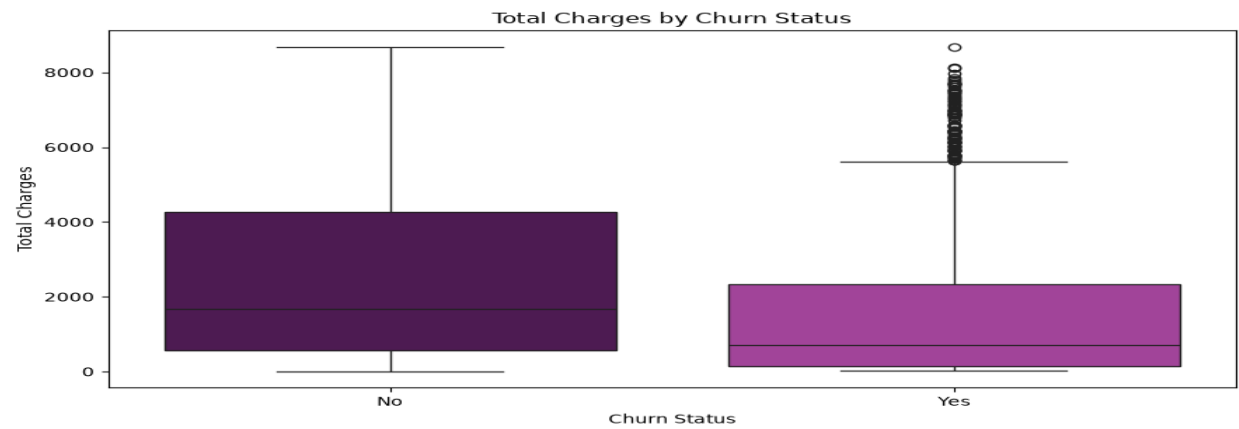
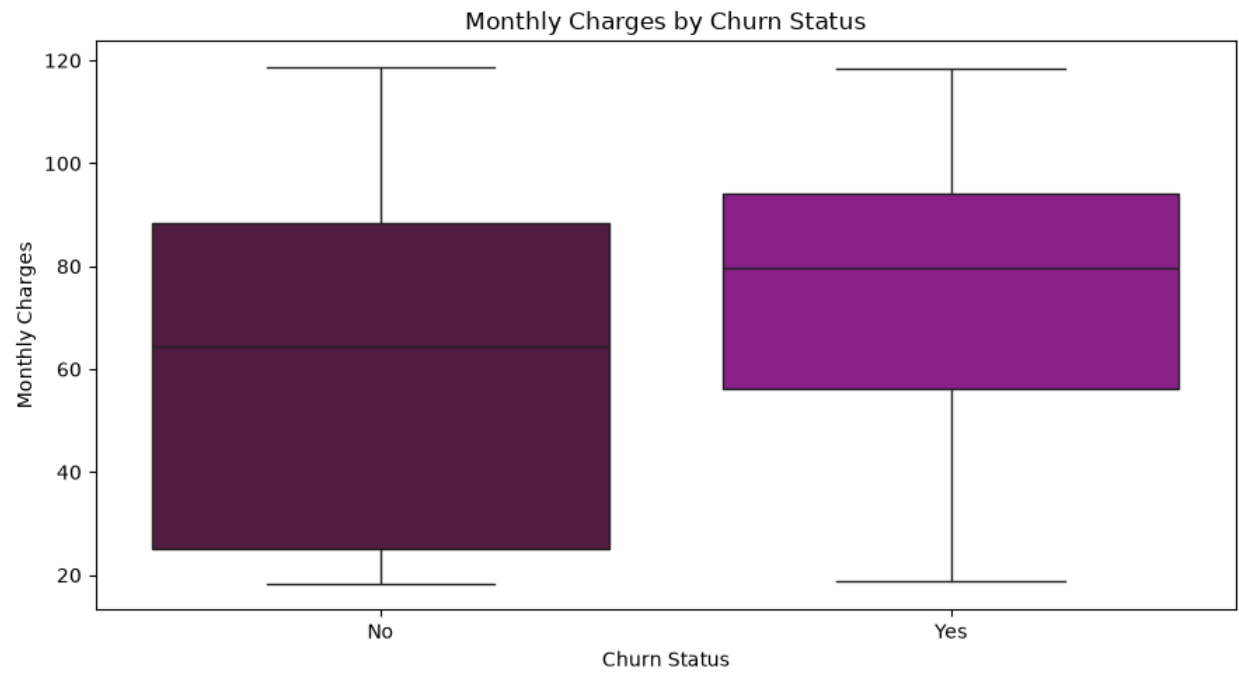
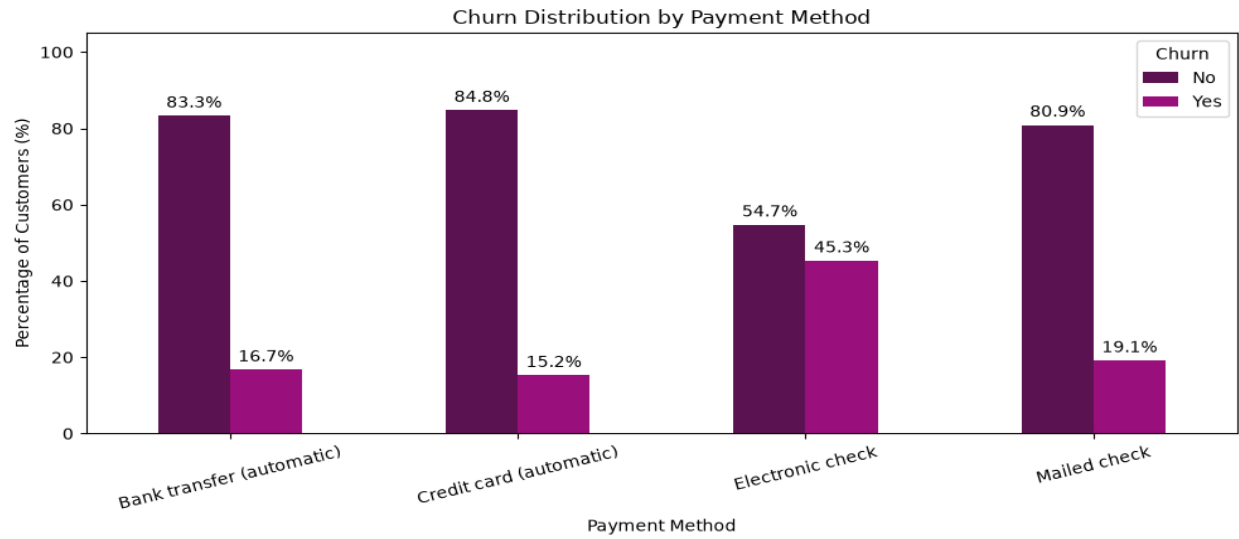
What these patterns suggested

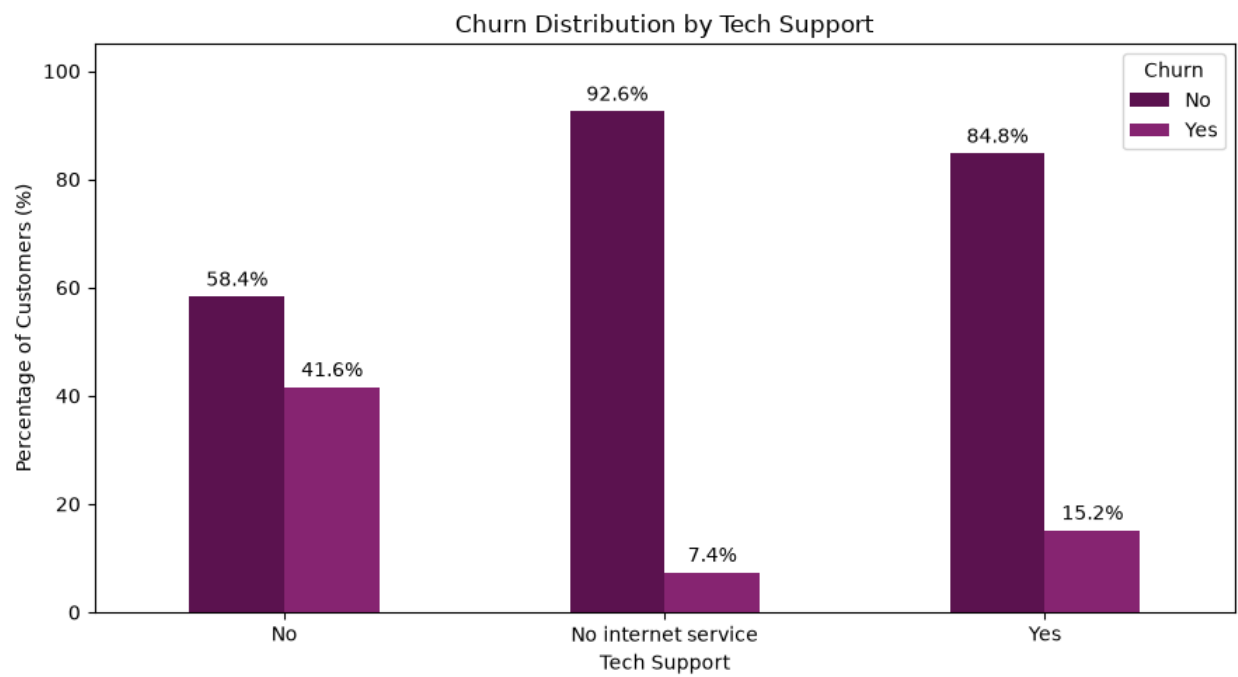
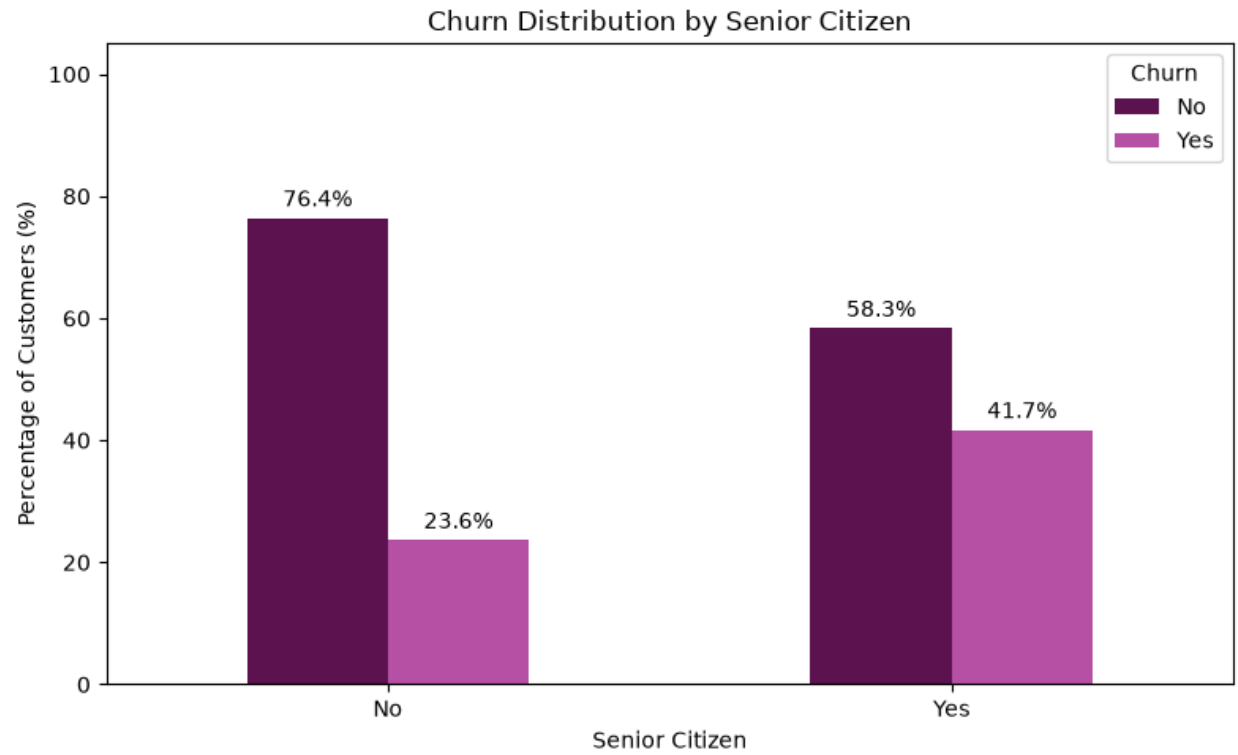
Every pattern above describes a binary outcome (did this customer churn or not) that needs to be predicted from a mix of numeric and categorical features with no natural table joins, no time-ordering to exploit, and no free-text fields to mine. That combination points directly at a supervised binary classification approach — the Predictive/Classification track — rather than Regression (no continuous target), Forecasting (no repeated time-series per customer), Segmentation (there is already a labeled outcome to predict, not an unlabeled grouping problem), or Text/Sentiment (no text fields in the data).

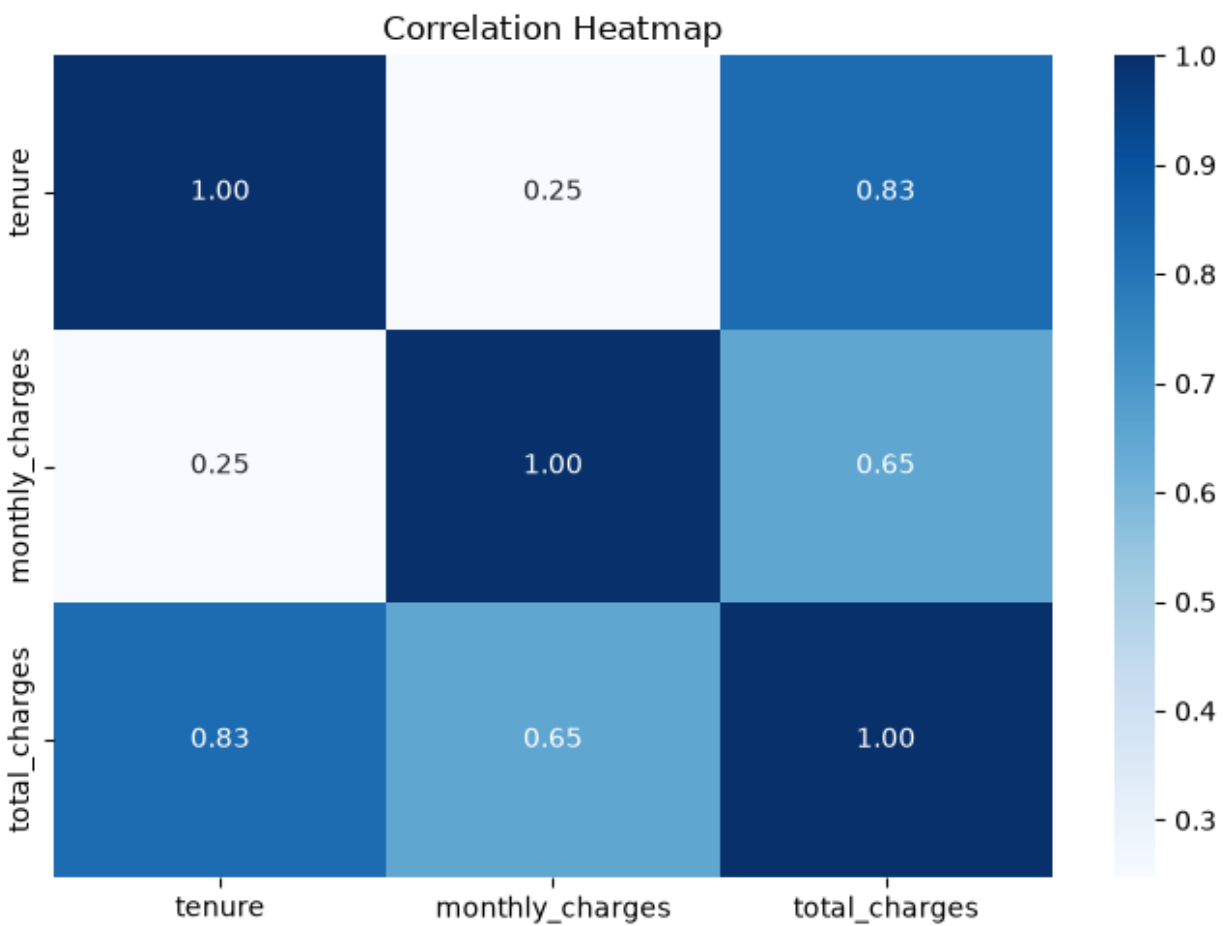
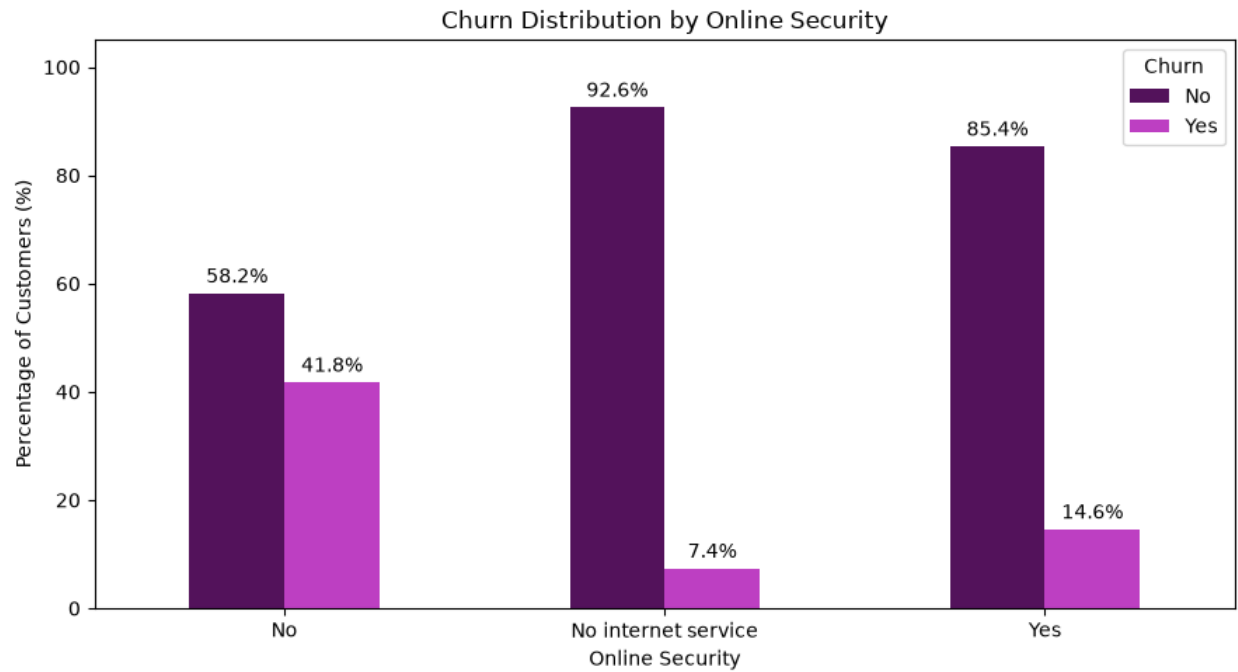
churn	No	Yes	Churn Rate in City
city			
Alexandria	1307	454	25.78%
Cairo	989	377	27.6%
Giza	444	162	26.73%
Banha	321	111	25.69%
Tanta	308	109	26.14%
Faiyum	314	100	24.15%
Damietta	235	95	28.79%
Mansoura	235	89	27.47%
Zagazig	244	86	26.06%
Sharm El Sheikh	219	69	23.96%
Faiyoun	149	67	31.02%
Suez	159	57	26.39%
El Mahalla El Kubra	143	55	27.78%
Port Said	106	38	26.39%

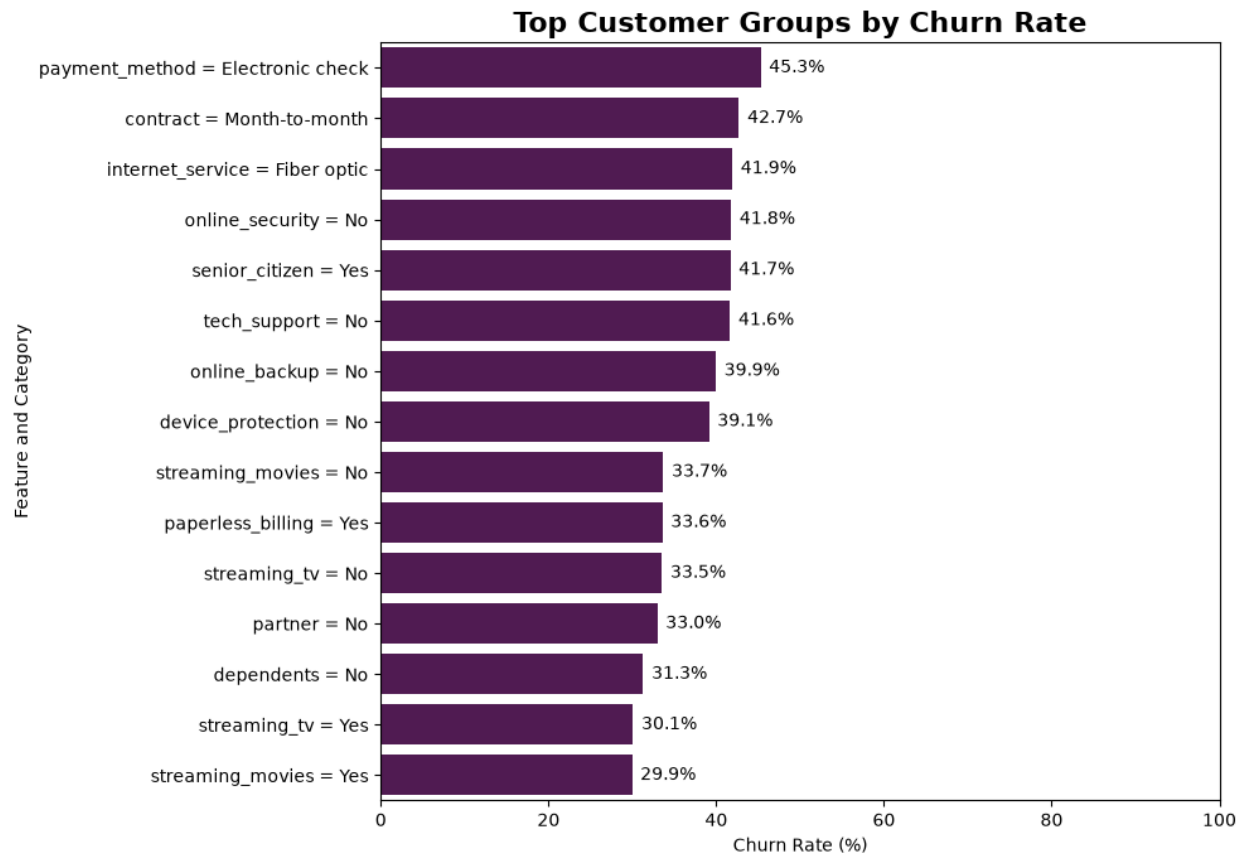












7. Technique Application — Modeling

Track chosen: Predictive / Classification

The target variable (churn) is binary (Yes/No), decided per-customer from a mix of demographic, service, and billing features — the textbook case for supervised classification. This fits the question better than the other tracks available: Regression requires a continuous target (churn is categorical); Forecasting requires repeated time-ordered observations per customer (this dataset is a single snapshot); Segmentation would group customers without using the known churn label, discarding the most useful information available; and Text/Sentiment does not apply, as there are no free-text fields.

Feature preparation

- Dropped customer_id (identifier, no predictive value) and tenure_group (redundant with tenure).
- Standardized the city field (merged the "Faiyum"/"Faiyoum" spelling duplicate) before encoding, so it contributes one clean signal instead of two diluted ones.
- Numeric features (tenure, monthly_charges, total_charges) scaled with StandardScaler; categorical features (including city, contract, payment_method, and the service columns) encoded with One-Hot Encoding — both fit on the training set only, inside a scikit-learn ColumnTransformer, to avoid data leakage.

- Class imbalance (26.5% churn) handled with SMOTE inside the modeling pipeline, applied only to the training folds during cross-validation — never to the test set.

Models tried and final choice

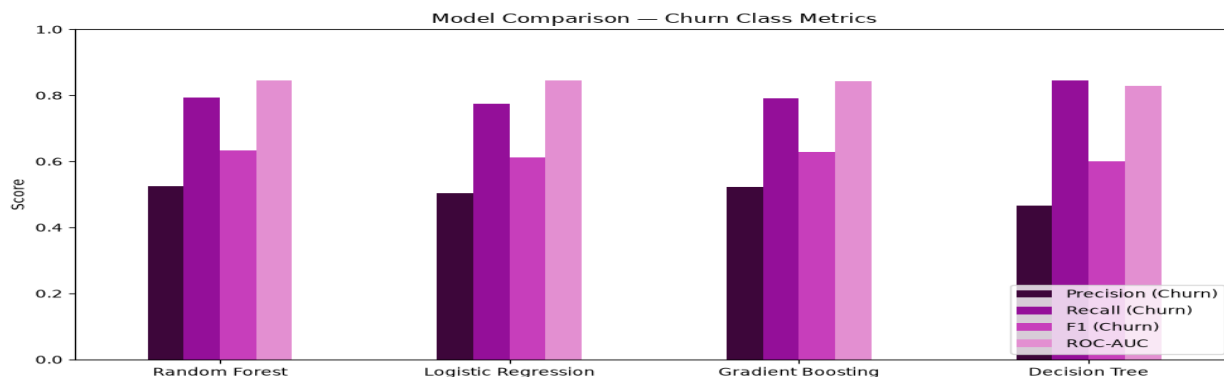
Five candidate models were compared using 5-fold Stratified Cross-Validation (to preserve the churn/no-churn ratio in every fold): Logistic Regression, Random Forest, Gradient Boosting, SVC, and K-Nearest Neighbors. The model with the best cross-validated F1-score was selected and then tuned with GridSearchCV.

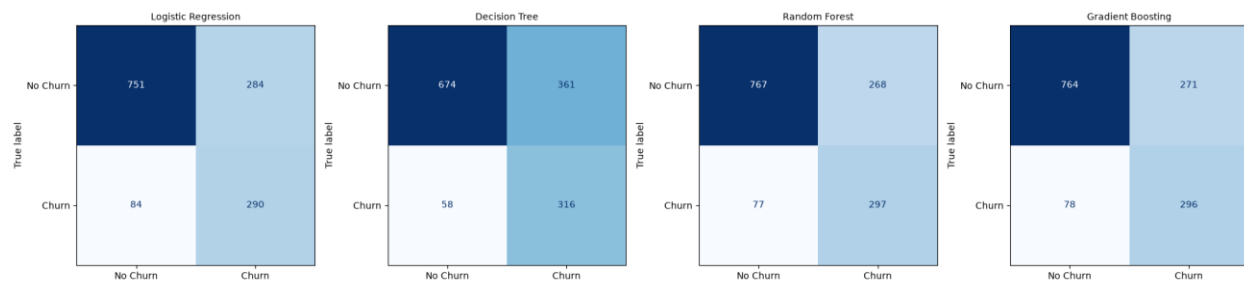
Model	F1 (CV)	Precision (CV)	Recall (CV)	ROC-AUC (CV)
Logistic Regression	0.612	0.505	0.775	0.845
Random Forest	0.633	0.526	0.794	0.846
Gradient Boosting	0.692	0.522	0.719	0.843
Decision Tree	0.601	0.476	0.845	0.830

[

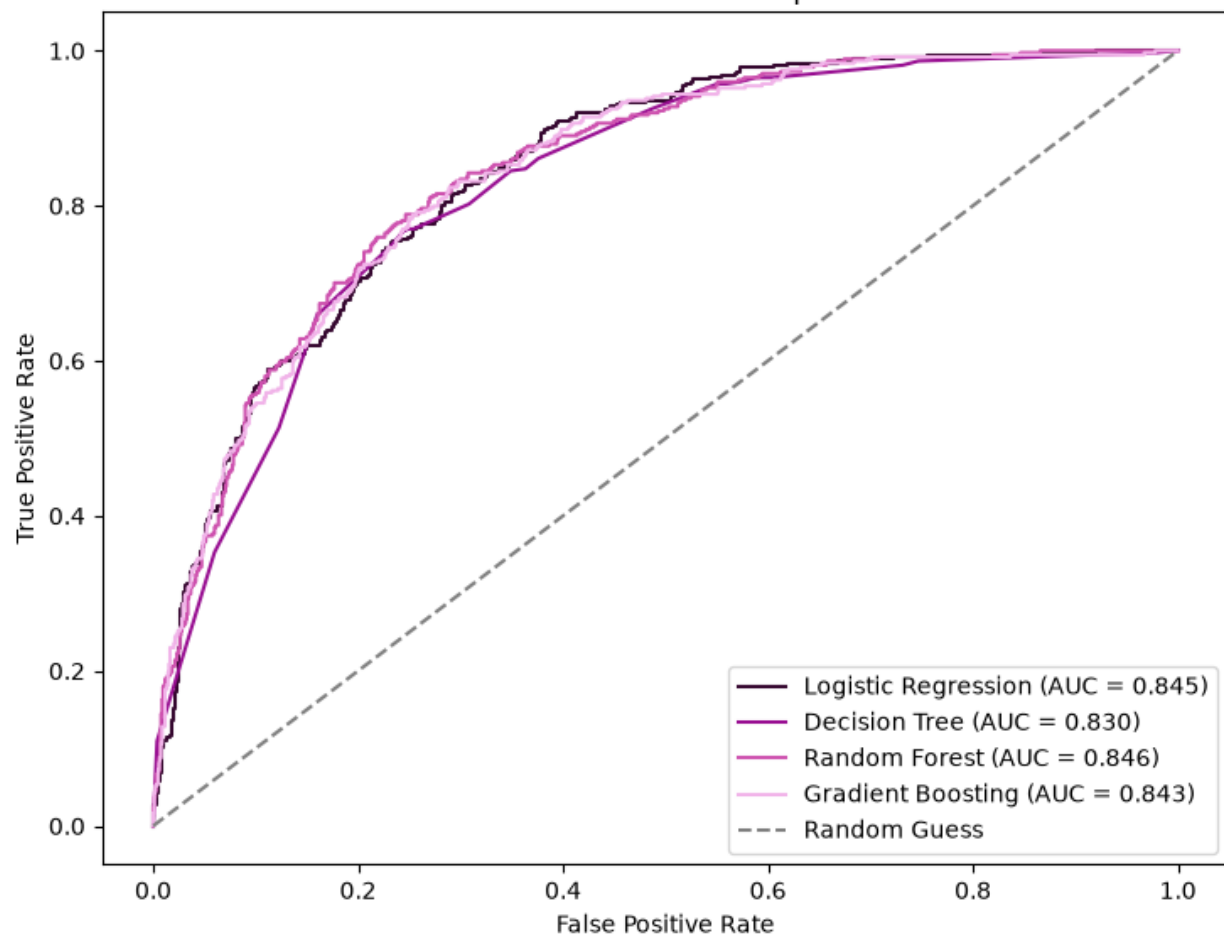
Final model evaluation (Logistic Regression)

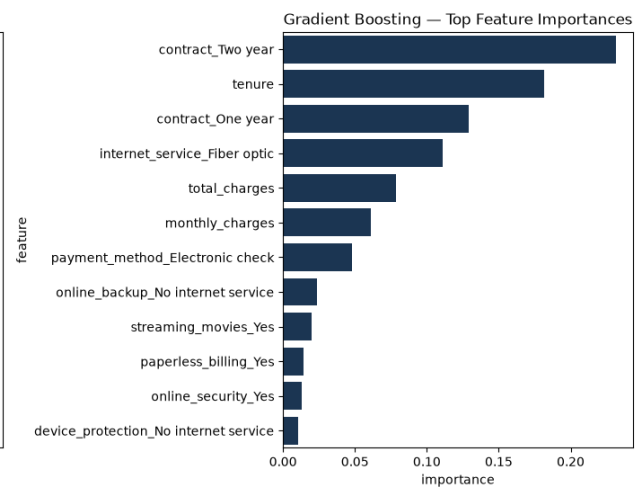
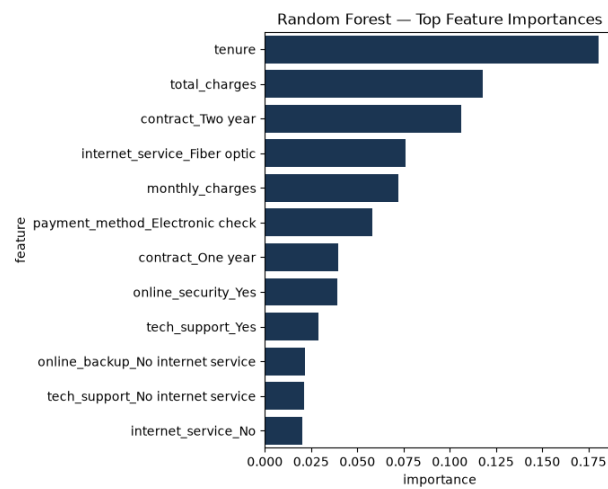
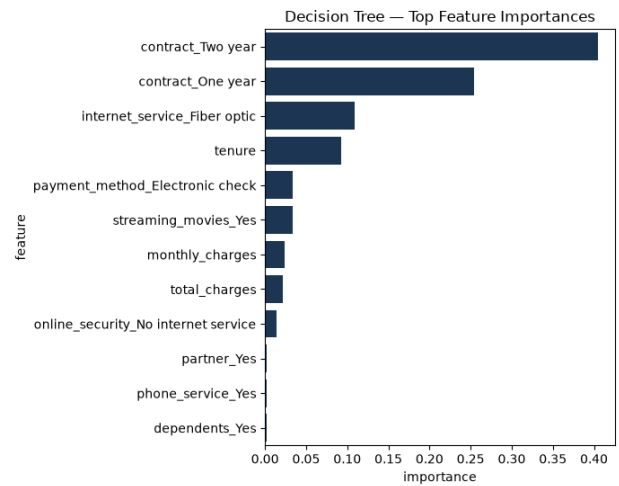
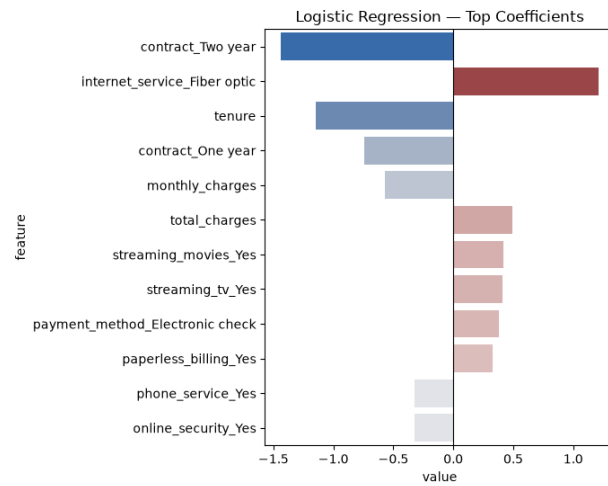
Metric	Value
Accuracy	0.739
Precision	0.505
Recall	0.775
F1-score	0.612
ROC-AUC	0.845

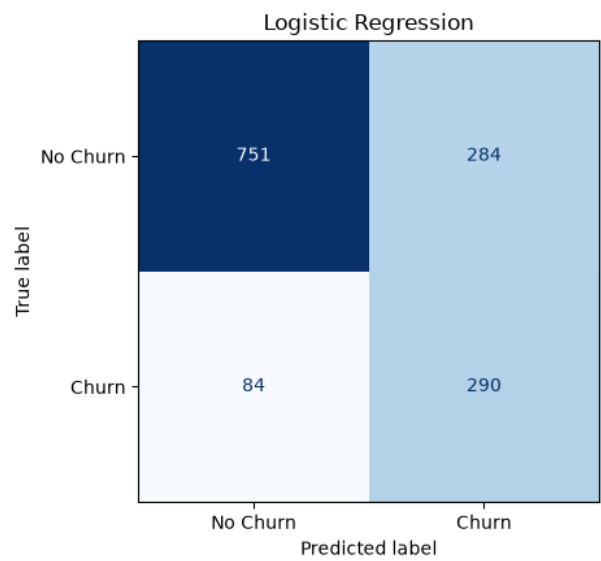
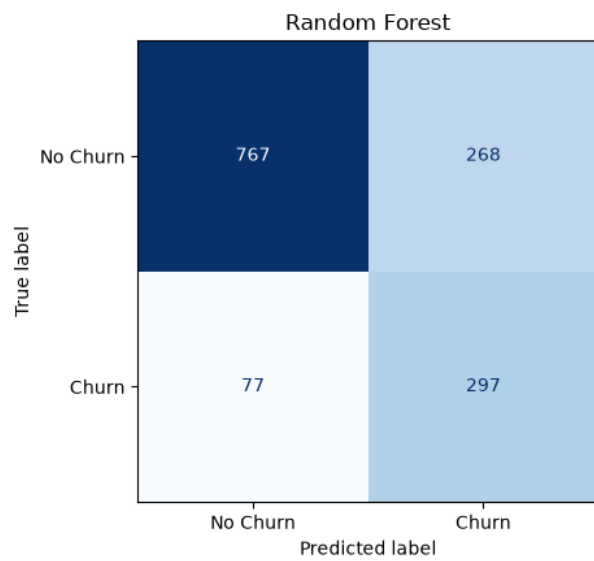
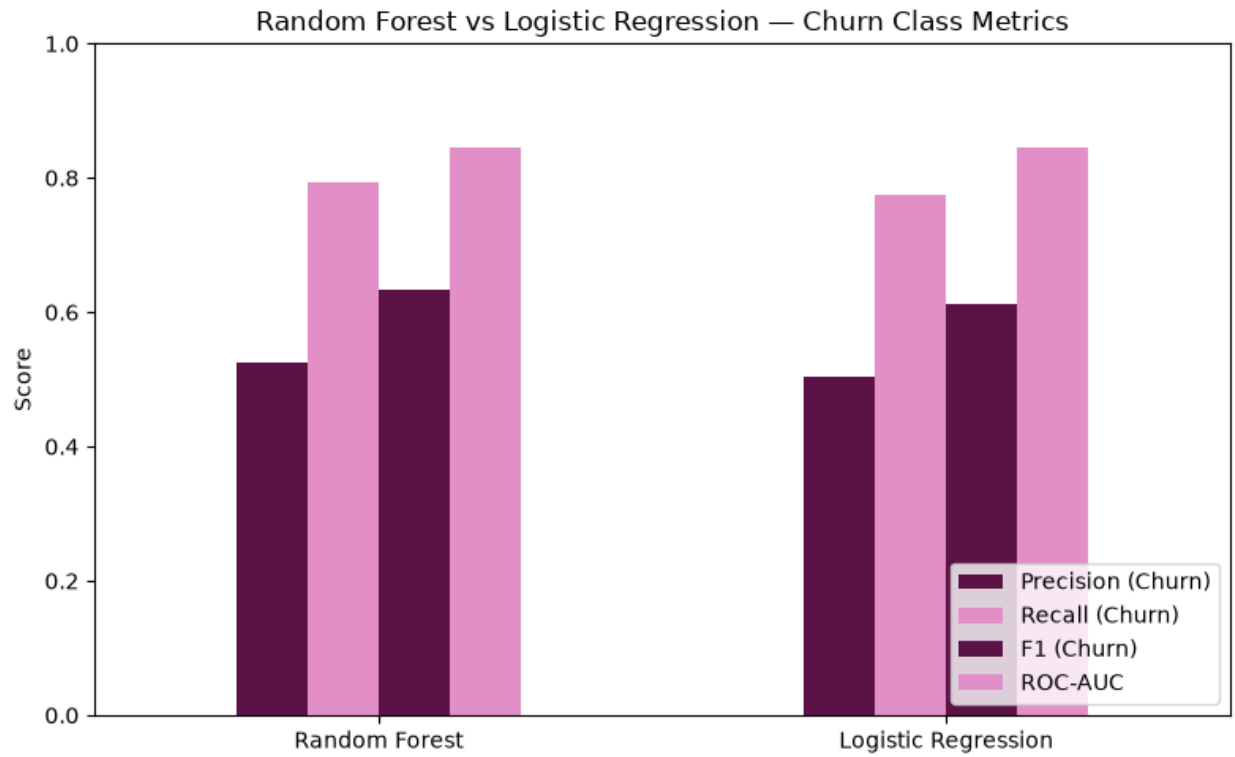


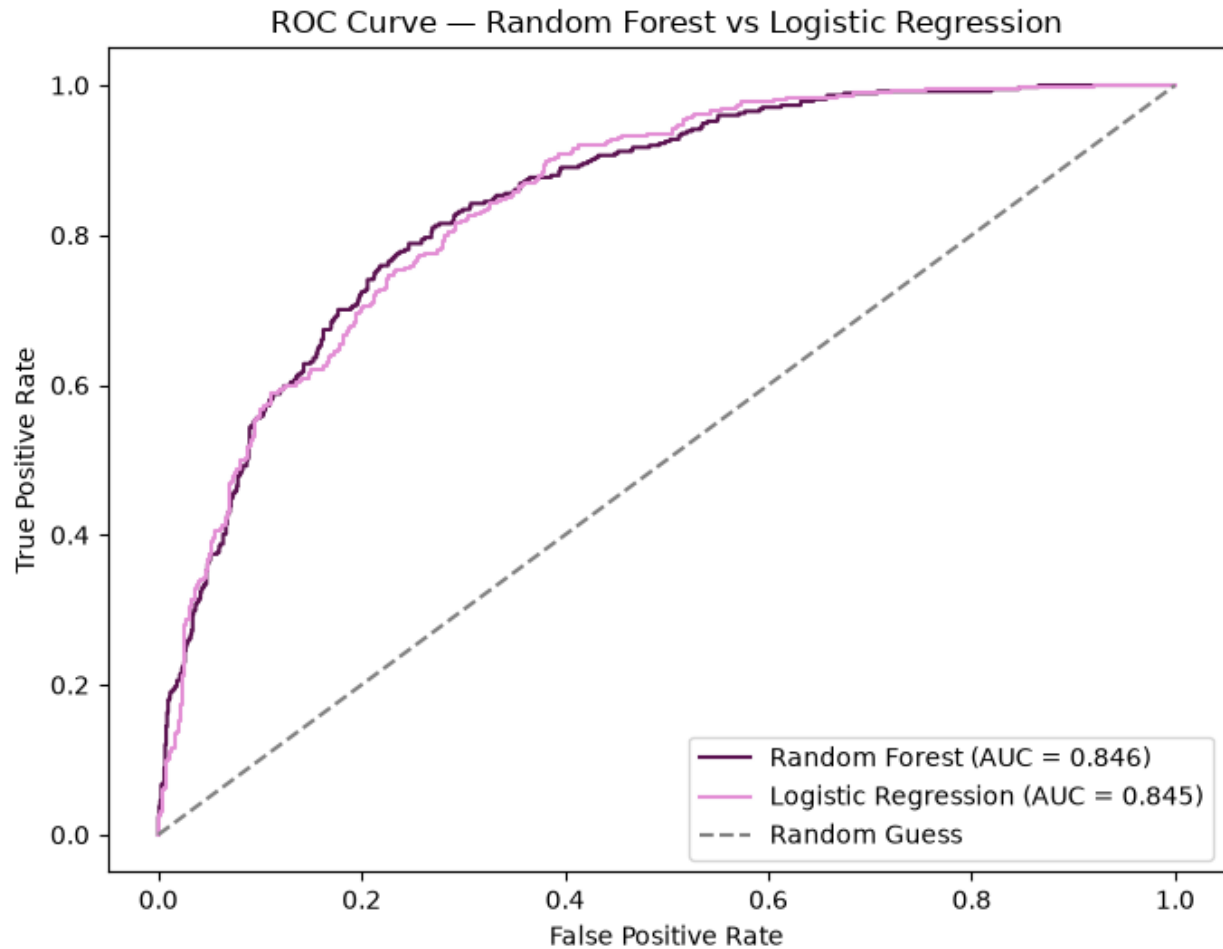


ROC Curve — Model Comparison









Streamlit app showcase

The final tuned pipeline was saved and wrapped in a Streamlit web app, so the trained model can be used without opening the notebook. A retention-team member enters a customer's account and service details (including the new city field) and receives an instant churn-risk score, with a clear high-risk / low-risk verdict.

Share

Telco Customer Churn Predictor

Model in use: Logistic Regression

Enter a customer's details below to estimate their probability of churning.

Customer Profile

City

Alexandria

Phone Service

Yes

Gender

Female

Multiple Lines

No

Senior Citizen

No

Internet Service

DSL

Has Partner

No

Online Security

No

تطبيق إدارة

No

No

Tenure (months)

12

Device Protection

No

Paperless Billing

No

Tech Support

No

Contract

Month-to-month

Streaming TV

No

Streaming Movies

No

Payment Method

Electronic check

Monthly Charges

70.00

Total Charges

1000.00

Predict Churn Probability

تطبيق إدارة

Monthly Charges

70.00

Total Charges

1000.00

Predict Churn Probability

تطبيق إدارة

Result

Predicted Churn Probability

48.3%

Medium risk — worth monitoring or a light-touch retention offer.

See the exact inputs used

تطبيق إدارة

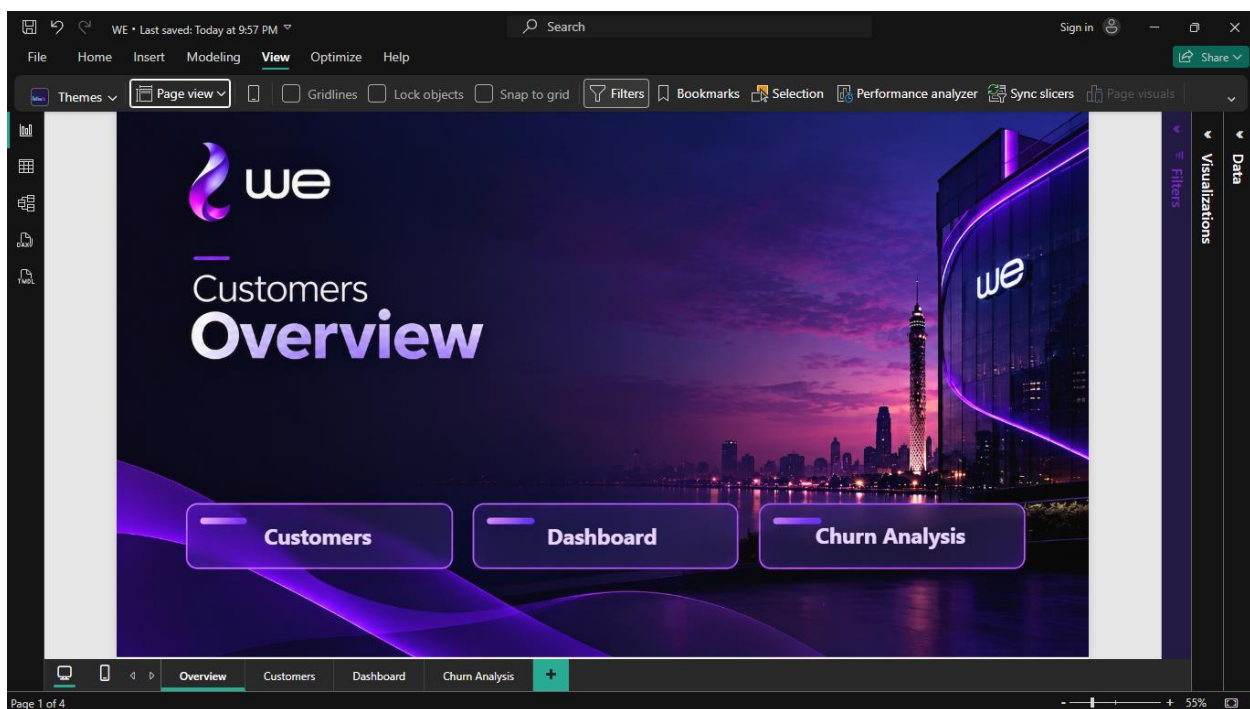
8. Dashboard & Inferred Insights

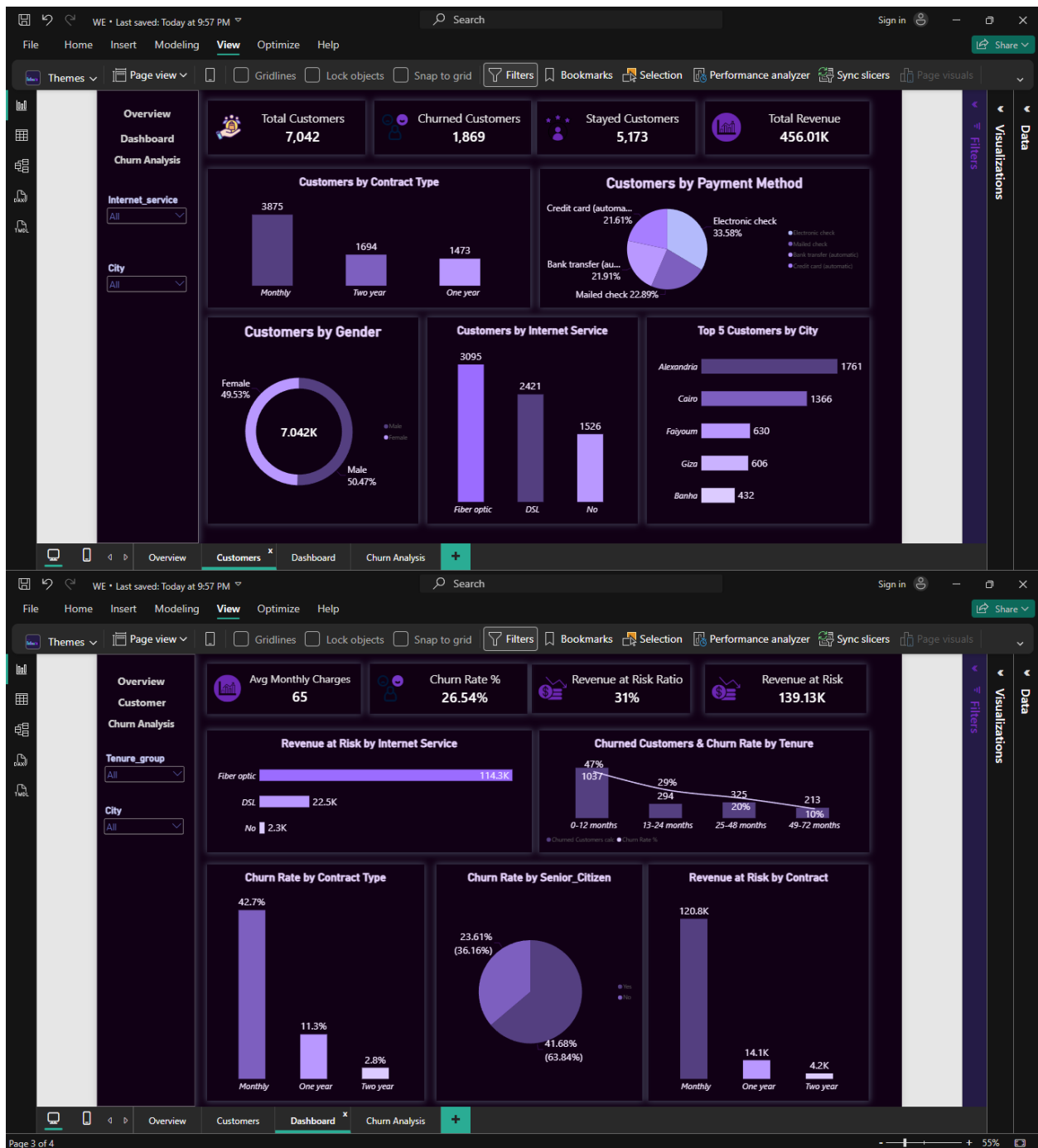
Dashboard structure

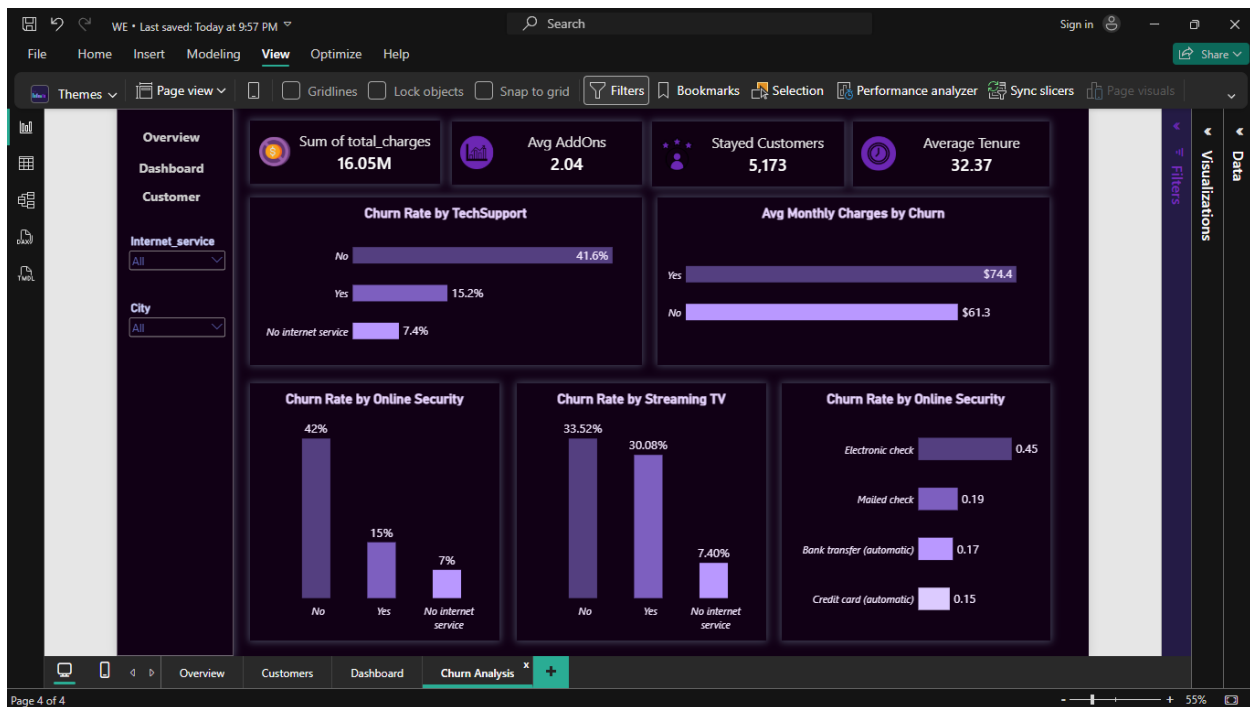
Page	What it answers
1. Overview	What is our overall churn rate, and how much revenue is at risk?
2. Demographics	Which customer segments (gender, age, household) churn more?
3. Services & Contract	How do contract type and payment method relate to churn?
4. Financial / Model Results	What do billing patterns and the model's risk scores tell us, and which features matter most?

Key insights

- Month-to-month contracts churn far more than one- or two-year contracts — the clearest actionable lever available (incentivize longer commitments).
- Customers paying by electronic check show the highest churn risk of any payment method.
- Shorter tenure and higher monthly charges both raise churn risk — newer, higher-paying customers need the most attention.
- Churn rate varies by city (~24%–31%) — worth a closer look at whether service quality or local competition differs by region.







9. Conclusion

Answer to the original question

Yes — churn can be predicted meaningfully ahead of time using the account, service, and billing data the company already holds. The trained model gives the retention team a ranked, actionable list of at-risk customers instead of waiting to observe cancellations after the fact.

Limitations

- The dataset is a single snapshot with no time dimension, so the model cannot yet detect a customer's risk changing month to month.
- The model's precision/recall trade-off ([insert your actual final numbers]) means some false positives (flagging a customer who would not have churned) and false negatives (missing a customer who does churn) are expected — retention outreach should be sized accordingly.
- Findings reflect this specific customer base and set of cities; applying the model elsewhere would need re-validation.

Recommendations / next steps

- Deploy the Streamlit app (or integrate the saved pipeline directly) so the retention team can score customers routinely, not just once.
- Prioritize retention offers for month-to-month, electronic-check customers in their first year — the segment with the clearest combined risk factors.

- Investigate the higher-churn cities directly (service quality, competitor presence) rather than relying on the model alone.
- Revisit the model periodically as new data accumulates, and consider adding a time dimension (e.g. monthly snapshots) to catch rising risk earlier.

10. Appendices

Links

- GitHub repository: [<https://github.com/sayedsawsan70-wq/Project-T-5#project-t-5>]
- Streamlit app: [<https://project-t-5-4plpxoevgxxnm8nuchhzbd.streamlit.app>]

Data source citation

Telco Customer Churn dataset. Source :

[<https://www.kaggle.com/code/basmalaawad/telco-customer-churn-dataset/notebook>].

-